



P.13 Which PWM node shall V_c connect to? How to know V_c value at steady-state?

解:

$$\frac{V_o}{V_g} = \frac{1}{1-D}$$

$$V_o \uparrow \rightarrow V_{FB} \uparrow$$

$$\rightarrow V_c \downarrow$$

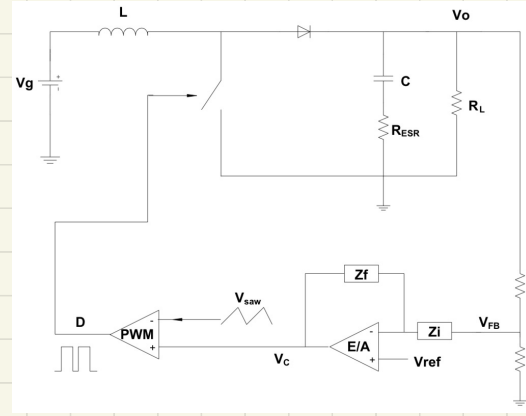
To make $V_o \downarrow$, $D \downarrow$

V_c connects to PWM's + node



$$\frac{V_c}{V_x} = D = 1 - \frac{V_g}{V_o}$$

$$V_c = V_x \left(1 - \frac{V_g}{V_o} \right)$$



PE-4- P. 16: Explain the control laws and why modulation waveform look like this

解

① peak control

d is set to 1 at CLK
when i_L reaches V_c , d back to 0

③ valley control

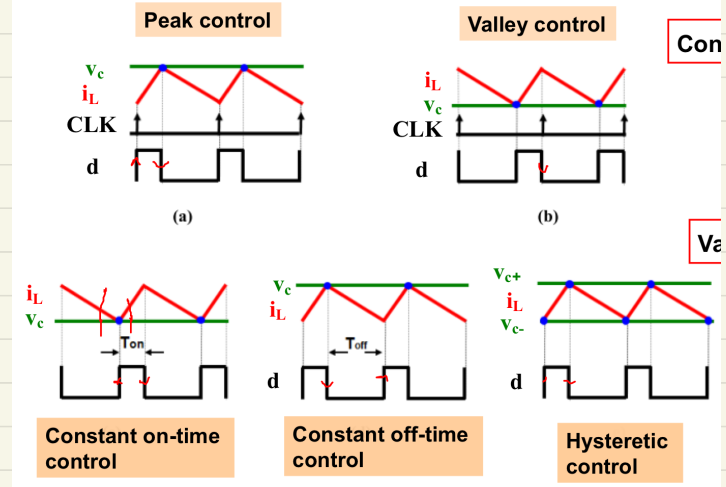
d is reset to 0 at CLK
when i_L reaches V_c , d is set to 1

③ constant on-time control

d is set to 1 when i_L reaches V_c
for a constant on time

④ Constant off-time control

d is set to 0 when i_L reaches V_c
for a constant off time



⑤ hysteresis control

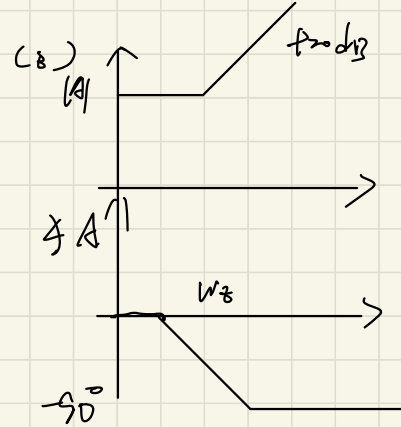
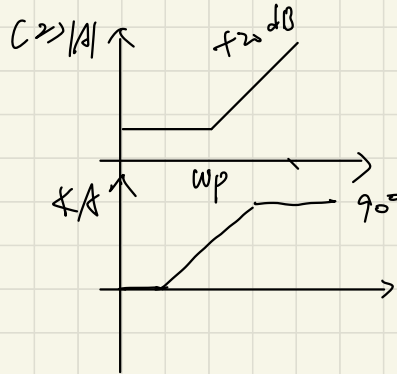
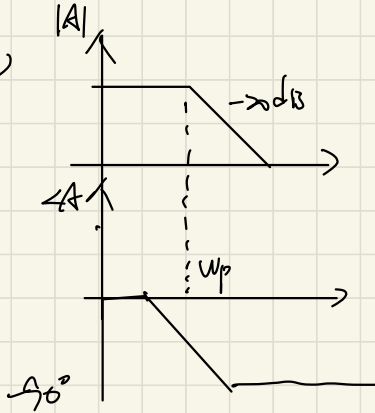
d turns on when i_L reaches V_{c-}
turns off when i_L reaches V_{c+}

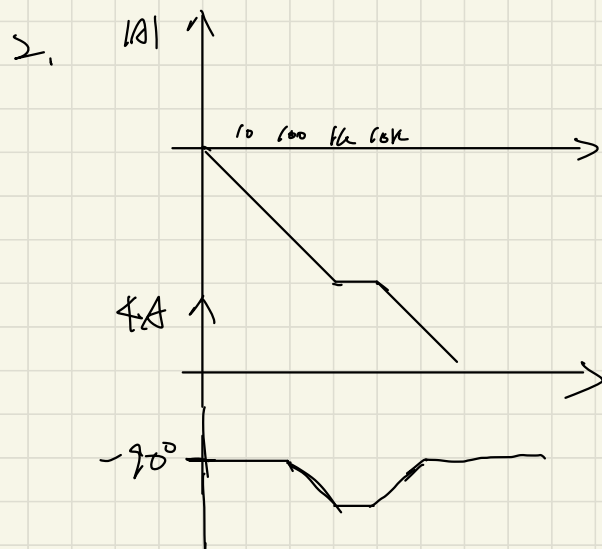
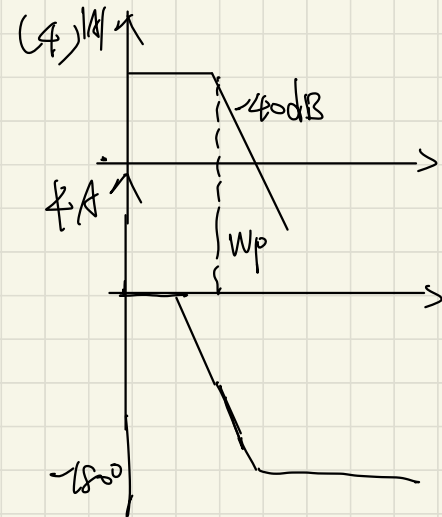
3, 1. Draw the Bode plot (gain and phase) of a transfer function with (1) single LHP pole, (2) LHP zero, (3) RHP zero, (4) a complex pole

2. Draw the Bode plot of transfer function with a pole at origin, a RHP zero at 1 KHz, a RHP pole at 10 KHz

Ans:

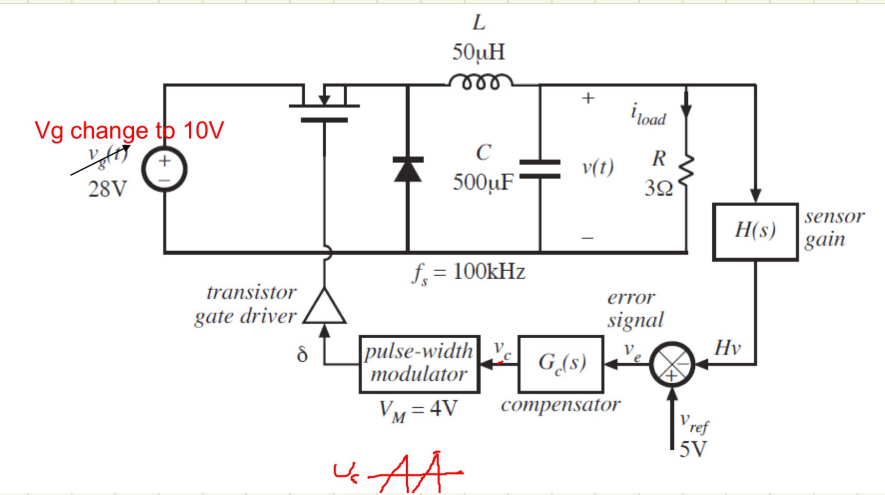
1. (1)





4.

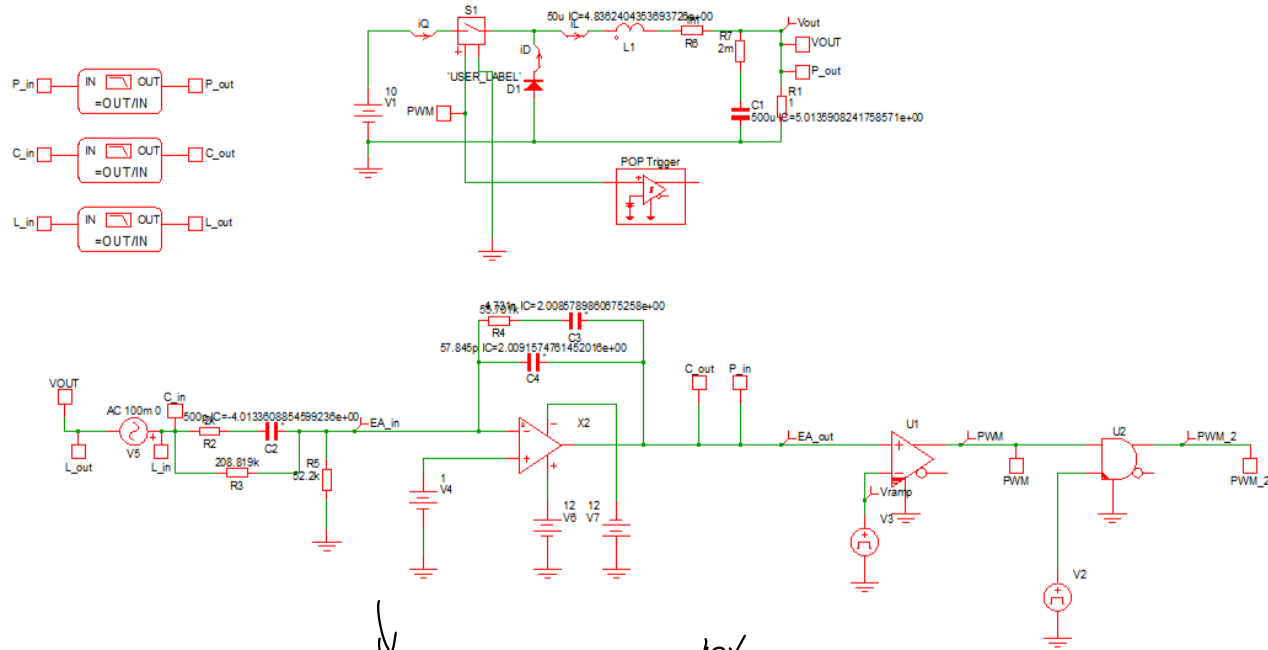
- Please design and implement a trailing-edge constant-frequency voltage-mode controlled buck converter.
- The spec. follows PEIC 6-p.16, 17 while **Vg change to 10V**, **Vout change to 5V**, **Vref change to 1V**, **R change to 1 ohm**.
The voltage of external ramp is between 1V and 5V.
- Design the compensator according to the uncompensated Loop Gain to achieve high loop gain bandwidth



- (1) Show the implemented close-loop behavior circuit in simulation software. SIMPLIS is recommended. Show the simulated steady-state waveforms of Vout and PWM comparator inputs/output signals. (10%)
- (2) Show how you design the compensator transfer function, compensation network values, and loop gain. Use the simulations of loop gain and compensator gain to verify your design. (50%)
- (3) If Vg has 100 mV sin. perturbation at 1 kHz, what is Vo perturbation magnitude before and after loop close? Use simulation to verify your calculation (10%)

32
(1)

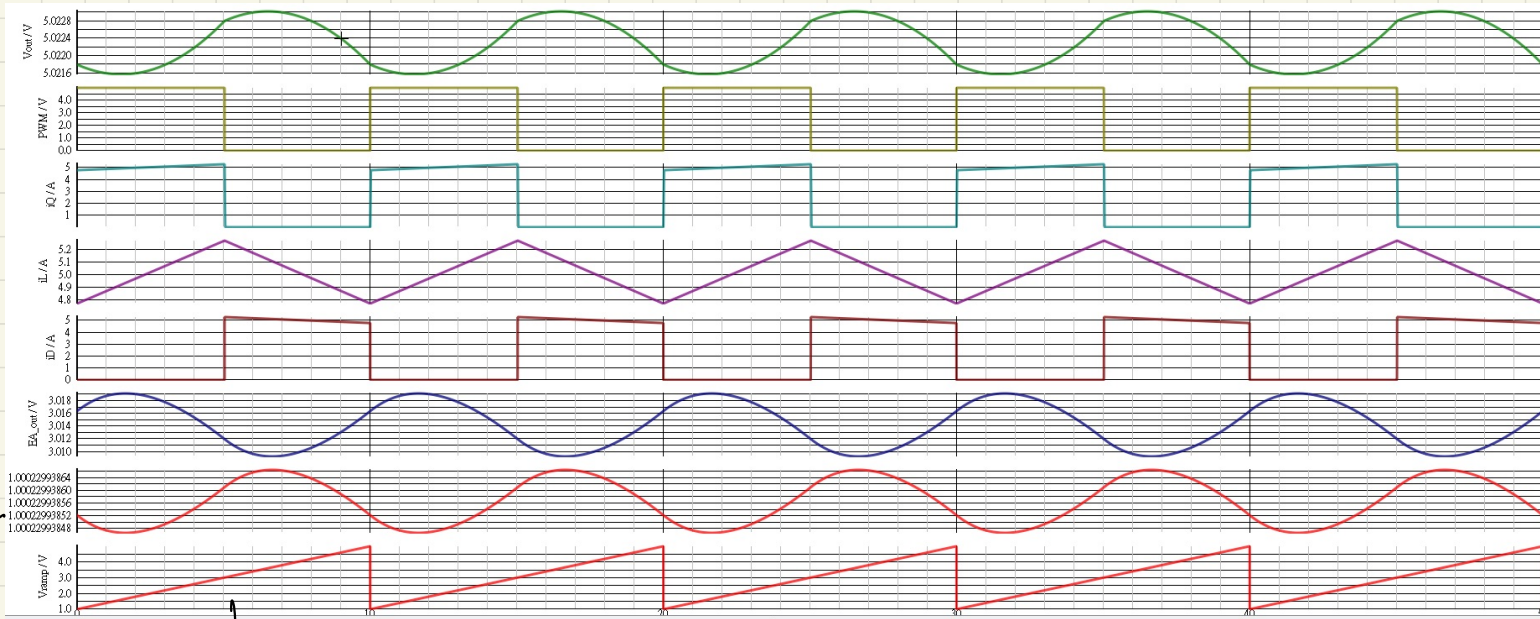
closed-loop behavior circuit



↓
Type III compensator

Steady-state waveform

V_{out}
PWM



Error
Amplifier
output

Error
Amplifier
Input

V_{ramp}

(2)

```
# --- 3. 補償器 sys_comp (使用修正後的 ZPK) ---
w0 = 1 / np.sqrt(L * C)
wesr = 1 / (C * rC)
wp1 = wesr
wz1, wz2 = 0.6 * w0, 1.5 * w0
wp2 = 0.5 * (2 * np.pi * fsw)
Kv = 1000 # 你的假設值
```

```
R1 = 2000.0

# (1) C1: 來自 Wp1 = 1 / (R1 * C1)
C1 = 1 / (R1 * wp1)

# (2) R2: 來自 Wz2 = 1 / ((R1 + R2) * C1)
R2 = (1 / (wz2 * C1)) - R1

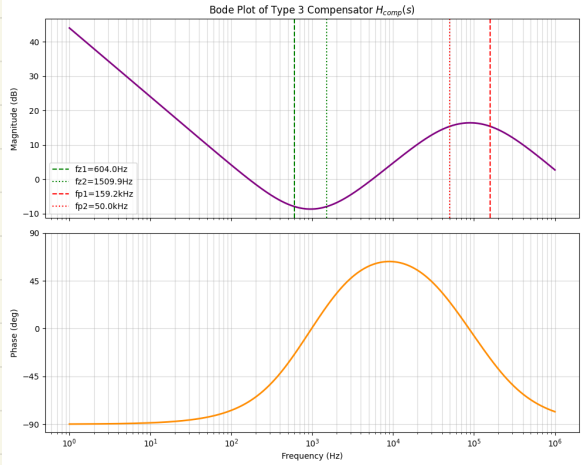
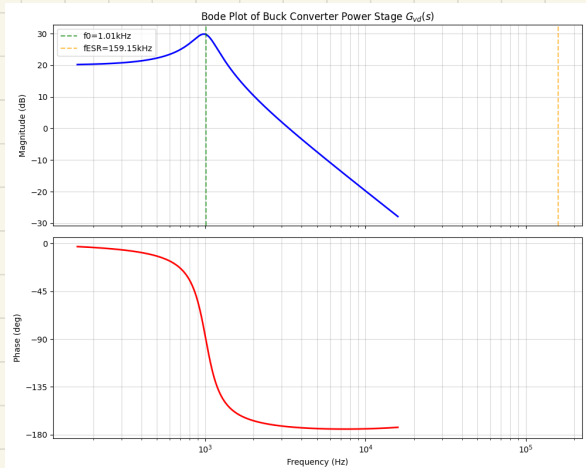
# (3) C2 & C3: 來自 Kv = 1 / (R2 * (C2 + C3)) 與 Wp2/Wz1 的關係
# 定義 C_sum = C2 + C3
C_sum = 1 / (Kv * R2)
# 根據 Wp2 = 1 / (R3 * (C2//C3)) 與 Wz1 = 1 / (R3 * C3)
# 可推導出 C2 = C_sum * (wz1 / wp2)
C2 = C_sum * (wz1 / wp2)
C3 = C_sum - C2

# (4) R3: 來自 Wz1 = 1 / (R3 * C3)
R3 = 1 / (wz1 * C3)
```

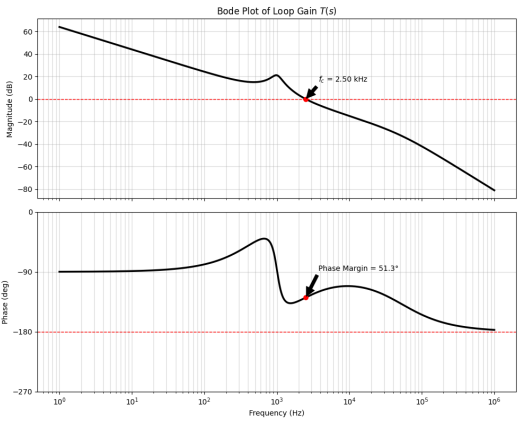
Crossover Frequency (fc): 2.50 kHz
Phase Margin (PM): 51.33 度

Key Frequency Values		
Parameter	rad/s	Hz
w0 (LC Resonance)	6.32e+03	1.01 kHz
wesr (Output Capacitor ESR)	1.00e+06	159.15 kHz
wz1 (Compensator Zero 1)	3.79e+03	603.95 Hz
wz2 (Compensator Zero 2)	9.49e+03	1.51 kHz
wp1 (Compensator Pole 1)	1.00e+06	159.15 kHz
wp2 (Compensator Pole 2)	3.14e+05	50.00 kHz

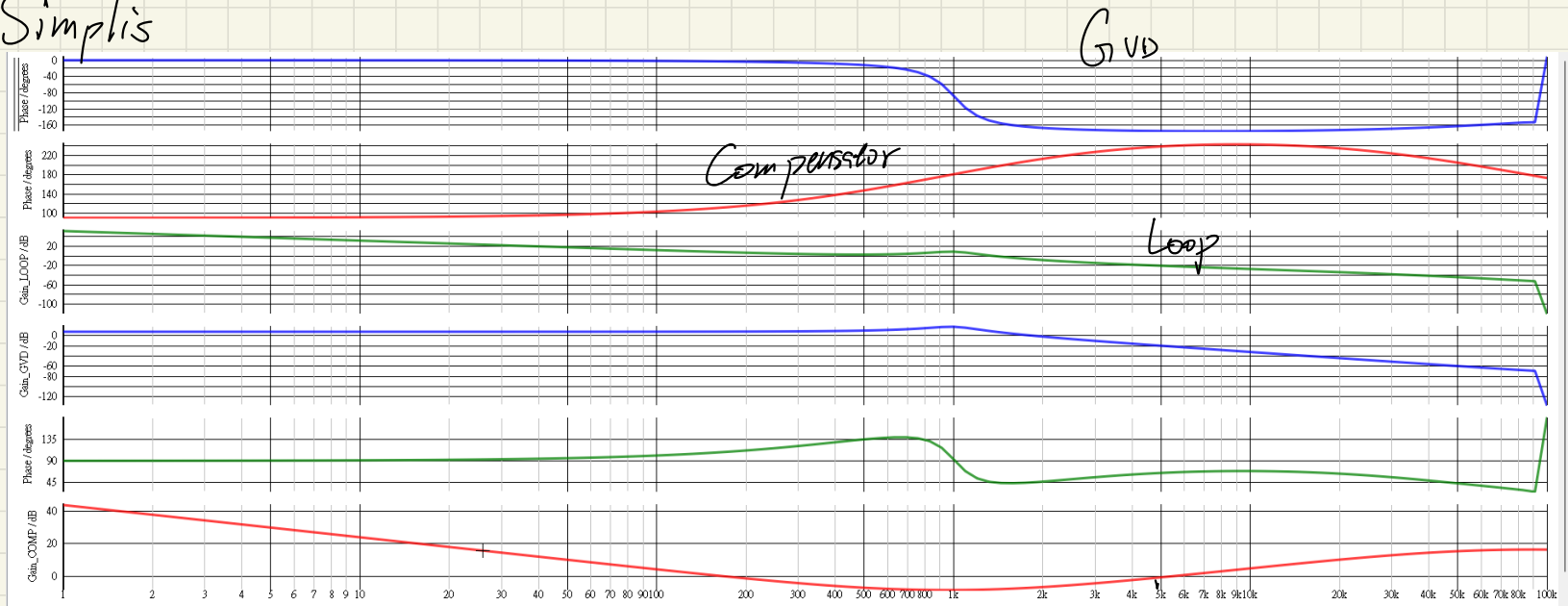
Type 3 Component Values	
R1 = 2.000 kOhm	
R2 = 208.819 kOhm	
R3 = 55.701 kOhm	
C1 = 500.000 pF	
C2 = 57.845 pF	
C3 = 4.731 nF	

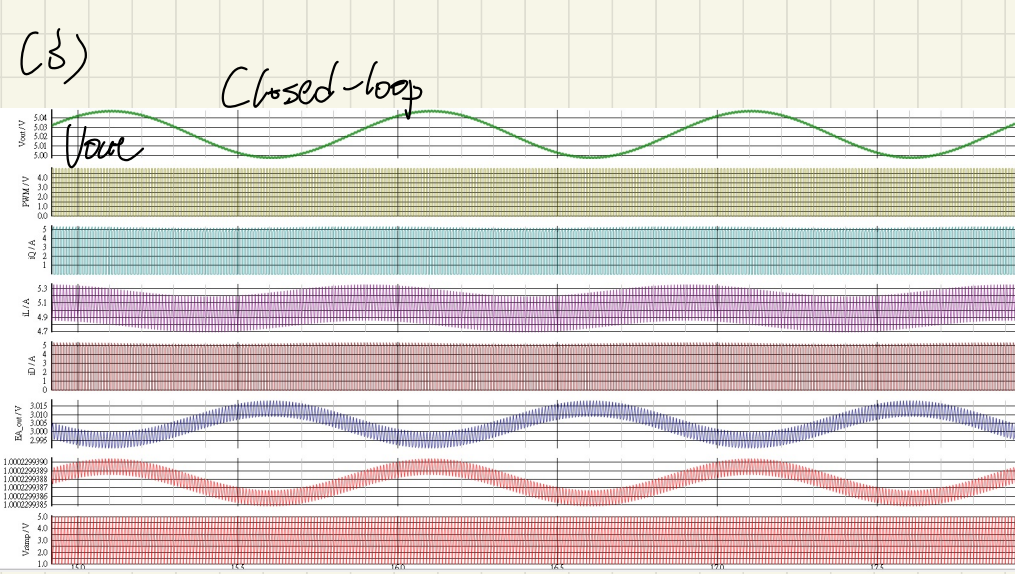


Python

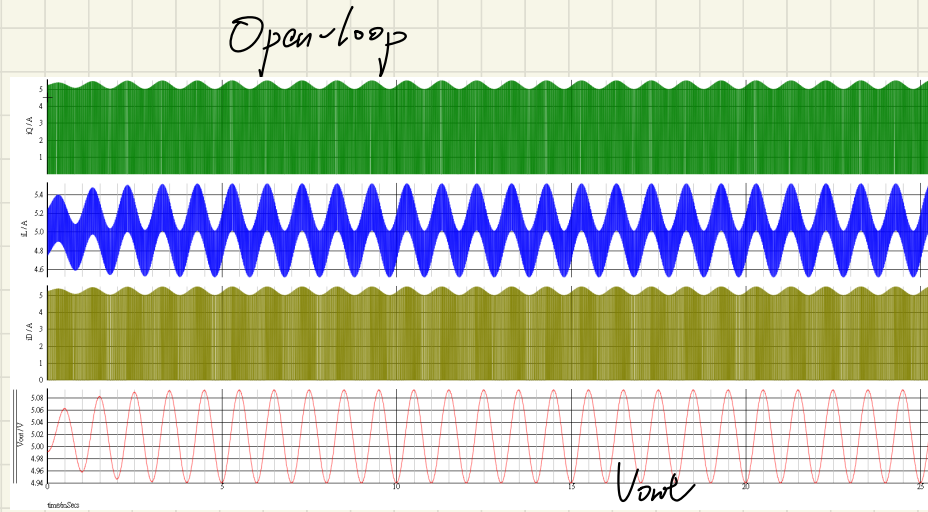


Simplis





Curve label	Name	Value
Vout	Peak To Peak	51.425067mV



Curve label	Name	Value
Vout	Peak To Peak	154.35651mV